Moments and reactions

National 5 · Mechanisms & structures · Answers

Answers and marking guidance

Working is shown for each answer. Other valid methods should be credited.

1a) Draw the free body diagram.
- Beam with 23 N at 1.8 m, 36 N at 4.0 m, 64 N at 6.3 m; RA at 0 and RB at 8.7 m

1b) Calculate, by taking moments about support A, the reaction at support B.
- RB × 8.7 = 23 × 1.8 + 36 × 4.0 + 64 × 6.3 = 588.6
- RB = 67.7 N

1c) Determine the reaction at support A.
- RA = 123 − 67.7 = 55.3 N

2a) Draw the free body diagram.
- 931 N at 9 m; reactions at 0 m and 25 m

2b) Calculate both reactions.
- W = 95 × 9.8 = 931 N
- Right: R × 25 = 931 × 9, R = 335 N
- Left: 931 − 335 = 596 N

3a) Draw the free body diagram.
- RA at 0, 15 kN at 25 m, 500 kN at 50 m, RB at 100 m

3b) Calculate RB and RA.
- RB × 100 = 15 × 25 + 500 × 50 = 25 375
- RB = 254 kN
- RA = 515 − 254 = 261 kN

4a) Draw the free body diagram.
- 500 N down, 350 mm from the right wheel (150 mm from the left); R1 at the left wheel, R2 at the right

4b) Calculate, by taking moments about R2, the reaction R1.
- R1 × 500 = 500 × 350
- R1 = 350 N

4c) Determine the reaction R2.
- R2 = 500 − 350 = 150 N

5a) Calculate the weight of the container.
- W = 500 × 9.8 = 4900 N

5b) Draw the free body diagram, showing all forces and distances.
- 4900 N at 127 from R2; span 587 between R1 and R2

5c) Calculate, by taking moments about R2, the reaction R1.
- R1 × 587 = 4900 × 127
- R1 = 1060 N

5d) Determine the reaction R2.
- R2 = 4900 − 1060 = 3840 N

6a) Draw the free body diagram.
- RA at 0, 750 N at 1.25 m, 1 kN at 1.5 m, RB at 3 m

6b) Calculate RA by taking moments about RB.
- RA × 3 = 750 × 1.75 + 1000 × 1.5 = 2812.5
- RA = 938 N

6c) Calculate RB.
- RB = 1750 − 938 = 813 N

7a) Draw the free body diagram.
- RA at 0, 100 N at 0.5 m, 350 N at 1.5 m, RB at 2.0 m

7b) Calculate RA by taking moments about RB.
- RA × 2 = 100 × 1.5 + 350 × 0.5 = 325
- RA = 163 N

7c) Calculate RB.
- RB = 450 − 163 = 288 N

8a)
- R2 = 100 × 1 ÷ 4 = 25 N
- R1 = 75 N

8b)
- R2 = (100 × 1 + 200 × 2) ÷ 4 = 125 N
- R1 = 175 N

8c)
- R2 = (60 × 1 + 40 × 2) ÷ 3 = 46.7 N (the 80 N acts at R1)
- R1 = 180 − 46.7 = 133 N

8d)
- R2 × 2 = 200 × 0.5 + 200 × 2.5 = 600, so R2 = 300 N
- R1 = 400 − 300 = 100 N

8e)
- R2 × 3 = 75 × 1 + 105 × 2 + 45 × 3.5 = 442.5, so R2 = 148 N
- R1 = 225 − 148 = 77.5 N

9a) Find R2 and the distance d.
- R2 = 100 − 40 = 60 N
- Moments about R1: 100 × d = 60 × 4, so d = 2.4 m

9b) Find the force F and the reaction R1.
- Moments about R1: F × 0.5 + 40 × 1.0 = 60 × 1.5, so F = 100 N
- R1 = 140 − 60 = 80 N

9c) Find the distance d.
- Moments about the left support: 1.5 × 1 + 2 × 1.75 + 0.7 × (3 + d) = 2.5 × 3
- 7.1 + 0.7d = 7.5, so d = 0.57 m (570 mm)